# Gravitational field strength

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a gravitational field to predict the force on a test mass.

**SUCCESS CRITERIA - I can...**

1. Represent field direction with arrows.
2. Use g=F/m and F=mg consistently.
3. Distinguish source field from the mass placed in it.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “field” and “gravity” to explain your answer. Sentence starters are optional.

**field**: a description of an influence at each location

**gravity**: an attractive interaction between masses

**radius**: distance from the center to the edge of a circle or sphere

## 01 Warm-up: bring back an earlier idea

An object weighs **29.4 N** where g=**9.8 N/kg**. **Find its mass** and its weight in a field of **5 N/kg**.

**Word help:** weight = the gravitational force on an object

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the prediction

**Reference:** $g = F_g/m = GM/r^2$.
Outside a spherical source of mass M; r is distance from its center. Field direction is toward the source.

**Worked reasoning:** A **2 kg** test object feels **6 N** toward a source. The local field is g=**6/2**=**3 N/kg**. A **4 kg** test object at that point feels **12 N**.

**Your turn:** 1. **Draw the field arrow** and the two possible force arrows.
2. Which quantity belongs to the location rather than the test mass?

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .

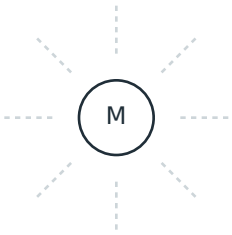
| Test mass (kg) | Force (N) | g (N/kg) |
|---|---|---|
| 2 | 6 | ___ |
| 4 | 12 | ___ |

**Marzano check - circle:** ① ② ③

## 03 Read and explain the diagram or equation

1. **Draw field arrows** at three locations around the source.
2. Which way would a small test mass accelerate?

**Sentence starter:** In my work, I used ... to show ... . I labeled it ... .




Add arrowheads to show field direction.

The center marks the source; add arrowheads on the surrounding lines to show field direction.

**Marzano check - circle:** ① ② ③

# Gravitational field strength

**antocciphysics**



Continue down each column.

Sentence starters are optional. You may plan in any language.

## 04 Apply the physics

A **0.20 kg** test mass experiences a **0.60 N** gravitational force. **Find the field strength**. The source has GM = **12 m³/s²**; **find the test mass's center-to-center** distance from the source.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




Source: GM = 12 Test mass: 0.20 kg

**Marzano check - circle:** 1 2 3

## 05 Explain your reasoning

A student says a heavier test mass falls faster because it feels more gravitational force. Use F=mg and a=F/m to respond.

**Sentence starter:** I notice ... . This supports ... because ... .

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

A **4 kg** block slides down a **30°** ramp with **11.6 N** friction up the ramp. Use g=**9.8 m/s²**. **Find along-ramp acceleration** and normal force.
**Draw the forces** on the chosen object and **label their sources**.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.

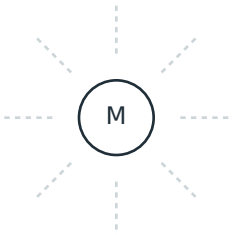
Add and label force arrows.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Outside a spherical source, field is **8 N/kg** at r. **Find field** at **2** r and force on **0.50 kg** there.

**Sentence starter:** I found ... [unit]. This value means ... in this situation.




Add arrowheads to show field direction.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the prediction

An exterior field is **8 N/kg** at distance r from a spherical source. Where is it **2 N/kg**? **What force** acts there on a **0.25 kg** mass?

**Sentence starter:** I notice ... . This supports ... because ... .

| Location | Test mass (kg) | Field (N/kg) | Force (N) |
|---|---|---|---|
| 1 | ____ | ____ | ____ |
| 2 | ____ | ____ | ____ |

**Marzano check - circle:** 1 2 3